

Канонические ур-я Гамильтона

$$X(x_1, \dots, x_n) : \det \left| \frac{\partial^2 X}{\partial x_i \partial x_j} \right| \neq 0$$

$$y_i = \frac{\partial X}{\partial x_i}$$

$$y_i = f_i(x_1, \dots, x_n) \quad i = \overline{1, n}$$

$$\Downarrow \leftarrow \text{если } \left| \frac{\partial f_i}{\partial x_j} \right| \neq 0$$

$$x_j = g_j(y_1, \dots, y_n) \quad j = \overline{1, n} \quad \text{если } \left| \frac{\partial^2 X}{\partial x_i \partial x_j} \right| \neq 0$$

Преобразование Лежандра от ф-и $X(x_1, \dots, x_n)$ к ф-и $Y(y_1, \dots, y_n)$:

$$Y = \sum_{i=1}^n x_i y_i - X(x_1, \dots, x_n)$$

$$x_i = x_i(y_1, \dots, y_n)$$

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Если преобраз. $X(x_1, \dots, x_n) : \det \left| \frac{\partial^2 X}{\partial x_i \partial x_j} \right| \neq 0$ и $\exists$ замена перемен-

ных $y_i = \frac{\partial X}{\partial x_i} \quad (1)$,

тогда $\exists$ обратная замена переменных $x_i = \frac{\partial Y}{\partial y_i} \quad (2)$, где ф-я Y обрат-

ной замены такая что $Y = \sum x_i y_i - X(x_1, \dots, x_n)$

Если кроме того $X(x_1, \dots, x_n, \alpha_1, \dots, \alpha_m)$, то ф-я $Y = Y(y_1, \dots, y_n, \alpha_1, \dots, \alpha_m)$

$$\text{и } \frac{\partial Y}{\partial \alpha_i} = - \frac{\partial X}{\partial \alpha_i}$$

$$\Delta. \quad y_i = \frac{\partial X}{\partial x_i} = f_i(x_1, \dots, x_n)$$

$$\Rightarrow \text{м.к. } \left| \frac{\partial^2 X}{\partial x_i \partial x_j} \right| \neq 0, \text{ то } x_i = g_i(y_1, \dots, y_n)$$

$$Y = \sum_{i=1}^n x_i y_i - X(x_1, \dots, x_n) = \sum_{i=1}^n g_i(y_1, \dots, y_n) y_i - X(g_1(\bar{y}), \dots, g_n(\bar{y}))$$

$$1) \quad \frac{\partial Y}{\partial y_i} = \sum_{j=1}^n \frac{\partial g_j}{\partial y_i} y_j + \sum_{j=1}^n g_j \frac{\partial y_j}{\partial y_i} - \sum_{j=1}^n \frac{\partial X}{\partial y_i} \frac{\partial y_j}{\partial y_i} = \sum_{j=1}^n \frac{\partial g_j}{\partial y_i} y_j + g_i(\bar{y}) - \sum_{j=1}^n \frac{\partial g_j}{\partial y_i} y_j = g_i(\bar{y}) = \underline{x_i}$$

$$2) \quad \exists X = X(x_1, \dots, x_n, \alpha_1, \dots, \alpha_m)$$

$$y_i = \frac{\partial X}{\partial x_i} = f_i(x_1, \dots, x_n, \alpha_1, \dots, \alpha_m)$$

$$Y = \sum x_i y_i - X(x_1, \dots, x_n, \alpha_1, \dots, \alpha_m)$$

$$x_j = g_j(y_1, \dots, y_n, \alpha_1, \dots, \alpha_m)$$

$$Y = \sum_{i=1}^n g_i(y_1, \dots, y_n, \alpha_1, \dots, \alpha_m) \cdot y_i - X(g_1(\bar{y}, \bar{\alpha}), \dots, g_n(\bar{y}, \bar{\alpha}), \alpha_1, \dots, \alpha_m)$$

$$\frac{\partial Y}{\partial \alpha_j} = \sum_{i=1}^n \frac{\partial g_i}{\partial \alpha_j} y_i - \sum_{i=1}^n \frac{\partial X}{\partial \alpha_j} \frac{\partial g_i}{\partial \alpha_j} - \frac{\partial X}{\partial \alpha_j} = - \frac{\partial X}{\partial \alpha_j}$$

$$\frac{\partial X}{\partial x_i} = y_i$$

$$x_1, \dots, x_n \rightarrow \dot{q}_1, \dots, \dot{q}_n$$

$$X \rightarrow L(\bar{q}, \dot{\bar{q}}, t)$$

$$y_i = \frac{\partial X}{\partial x_i} \Rightarrow \frac{\partial L}{\partial \dot{q}_i} = p_i$$

$$\dot{q}_i = \frac{\partial Y}{\partial p_i} \quad ? Y$$

$$Y \rightarrow H, \quad H = \sum p_i \dot{q}_i - L(\bar{q}, \dot{\bar{q}}, t)$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \rightarrow \text{1-е уравнение}$$

$$x_1, \dots, x_m \rightarrow q_1, \dots, q_n, t$$

$$\frac{\partial L}{\partial q_i} = - \frac{\partial H}{\partial q_i}$$

$$\frac{\partial L}{\partial t} = - \frac{\partial H}{\partial t}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \quad i = \overline{1, n}$$

$$\frac{d}{dt} p_i = \frac{\partial L}{\partial q_i} = - \frac{\partial H}{\partial q_i}$$

$$\dot{p}_i = - \frac{\partial H}{\partial q_i} \rightarrow \text{2-е уравнение}$$

Как перейти от L к H ?

$$1) p_i = \frac{\partial L}{\partial \dot{q}_i} \quad i = \overline{1, n}$$

$$2) H = \sum p_i \dot{q}_i - L(\bar{q}, \dot{\bar{q}}, t)$$

$$H(\bar{q}, \bar{p}, t)$$

$$\text{из (1)} \quad p_i = \frac{\partial L}{\partial \dot{q}_i} = f_i(\bar{q}, \bar{p}, t) \Rightarrow \text{если } \left| \frac{\partial f_i}{\partial \dot{q}_i} \right| \neq 0 \rightarrow \dot{q}_i = g_i(\bar{q}, \bar{p}, t)$$

$$\left| \frac{\partial f_i}{\partial \dot{q}_i} \right| = \left| \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_i} \right|$$

$$\text{если } L = T - \Pi \quad (\Pi \rightarrow V), \text{ то}$$

$$\left| \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_i} \right| \neq 0$$

$$\|a_{ij}\| \neq 0$$

$$3) \begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = - \frac{\partial H}{\partial q_i} \end{cases} \quad i = \overline{1, n}$$

уравнения Гамильтона

Как перейти от H к L ?

$$1) \dot{q}_i = \frac{\partial H}{\partial p_i} = h_i(\bar{q}, \bar{p}, t)$$

$$2) L = \sum p_i \dot{q}_i - H(\bar{q}, \bar{p}, t)$$

$$\text{из (1)} \quad p_i = p_i(\bar{q}, \dot{\bar{q}}, t), \text{ если } \left| \frac{\partial h_i}{\partial p_i} \right| \neq 0$$

$$\left| \frac{\partial^2 H}{\partial p_i \partial p_i} \right| \neq 0$$

Физический смысл ф-ы Н

$$H = \sum p_i \dot{q}_i - L(\bar{q}, \dot{\bar{q}}, t) = \sum \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L$$

↑ натур., с пом. П

$$L = T - \Pi = T_2 + T_1 + T_0 - \Pi$$

th теорема об одн. ф-ях

$f(x_1, \dots, x_n)$ - одн. степен. к, то

$$\sum_{i=1}^n \frac{\partial f}{\partial x_i} x_i = k f$$

$$\sum \frac{\partial T_2}{\partial \dot{q}_i} \dot{q}_i = 2 T_2$$

$$\sum \frac{\partial T_0}{\partial \dot{q}_i} \dot{q}_i = 0$$

$$\sum \frac{\partial T_1}{\partial \dot{q}_i} \dot{q}_i = 1 \cdot T_1$$

$$\sum \frac{\partial \Pi}{\partial \dot{q}_i} \dot{q}_i = 0$$

$$\sum \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i = 2 T_2 + T_1$$

$$H = 2 T_2 + T_1 - T_2 - T_1 - T_0 + \Pi = T_2 - T_0 + \Pi \quad \text{— обобщ. энергия}$$

Если сист. замкн. и стационарн., то $T_1 = T_0 = 0$ (м.к. $\frac{\partial \bar{r}}{\partial t} = 0$)

$$\Rightarrow H = T_2 + \Pi = T + \Pi$$

Первые интегралы ур-я Гамильтона

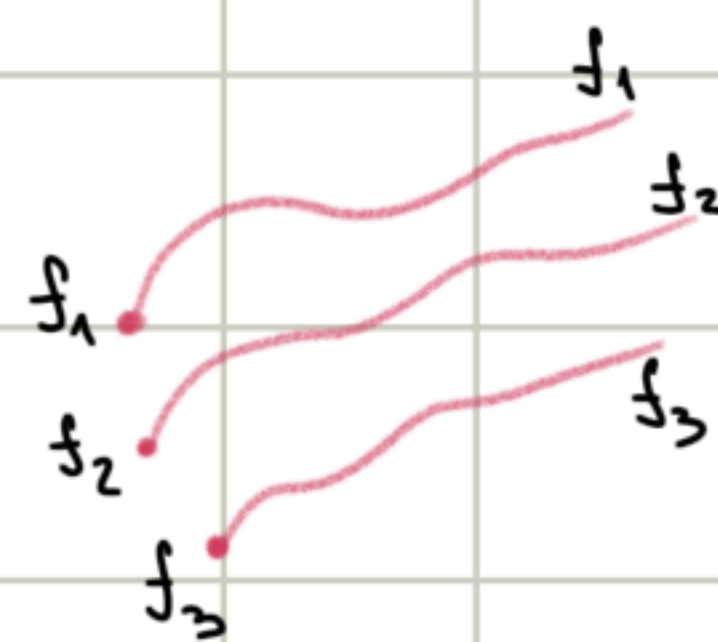
ф-я $f(\bar{q}, \bar{p}, t)$ - ПИ сист. диффузрв $\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = - \frac{\partial H}{\partial q_i} \end{cases}$, если

$f = \text{const}$ на траекториях сист.

$$f(\bar{q}(q_0, p_0, t), \bar{p}(q_0, p_0, t), t) = f(q_0, p_0, t_0)$$

$$\left. \frac{df}{dt} \right|_H = \sum \frac{\partial f}{\partial q_i} \dot{q}_i \Big|_H = \sum \frac{\partial f}{\partial p_i} \dot{p}_i \Big|_H + \frac{\partial f}{\partial t} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right) + \frac{\partial f}{\partial t}$$

$$f - \text{ПИ} \Leftrightarrow \left. \frac{df}{dt} \right|_H = 0$$



1) Циклическая координата

q_i - цикл., если $\frac{\partial L}{\partial q_i} = 0$

$$\frac{\partial H}{\partial q_i} = - \frac{\partial L}{\partial q_i} = 0$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \quad \Rightarrow \quad \frac{\partial L}{\partial \dot{q}_i} = \text{const}$$

$$\dot{p}_i = - \frac{\partial H}{\partial q_i} = 0 \quad \Rightarrow \quad p_i = \text{const}$$

если q_i - цикли, то p_i - ПУ

2) Обобщ. консер. сист.

$$\frac{\partial H}{\partial t} = 0 \quad H = H(\bar{q}, \bar{p})$$

$$\left. \frac{dH}{dt} \right|_H = \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} \frac{dq_i}{dt} - \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} \right) + \frac{\partial H}{\partial t}$$

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} \quad (\text{всегда!})$$

$$\text{а если } \frac{\partial H}{\partial t} = 0 \Rightarrow \frac{dH}{dt} = 0 \Rightarrow H - \text{ПУ}$$

обобщ. эн. - ПУ (интеграл энергии и Якоби)

$$H = T_2 - T_0 + \Pi = \text{const} \quad !$$

если сист. консерв. (склероном., $\frac{\partial \Pi}{\partial t} = 0$)

$$H = T + \Pi$$

$$\frac{\partial H}{\partial t} = 0 \Rightarrow T + \Pi = \text{const} \quad (\text{полн. мех. эн. const})$$

Понижение порядка баш. сист. с использованием ПУ

1) $\exists q_1$ - цикли. коорд.

$$H = H(q_2, \dots, q_n, p_2, \dots, p_n, c_1, t) \quad \swarrow p_1$$

$$i = \overline{2, n} \quad \begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} = f_i(q_2, \dots, q_n, p_2, \dots, p_n, c_1, t) \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} = g_i(q_2, \dots, q_n, p_2, \dots, p_n, c_1, t) \end{cases} \quad (*)$$

Узнаваемо $2n$ ур-н

если q_1 - цикли, то $(*)$ $2n-2$ ур-н

$$(*) \Rightarrow \begin{cases} q_i = q_i(c_1, c_2, c'_2, \dots, c_n, c'_n, t) \\ p_i = p_i(c_1, c_2, c'_2, \dots, c_n, c'_n, t) \end{cases} \quad \begin{matrix} i = \overline{2, n} \\ (**) \end{matrix}$$

$$\dot{q}_1 = \left. \frac{\partial H}{\partial p_1} \right|_{p_1=c_1} = h_1(c_1, c_2, c'_2, \dots, c_n, c'_n, t) \quad (**)$$

$$q_1 = \int h_1(\dots) dt + c'_1$$

2) Обобщ. консер. сист.

$$H = H(\bar{p}, \bar{q})$$

$$H(p, q) = h, \text{ т.к. } \frac{\partial H}{\partial t} = 0 \quad H - \text{ПУ}$$

$$\exists \frac{\partial H}{\partial p_1} \neq 0 \Rightarrow p_1 = -k(q_1, \dots, q_n, p_2, \dots, p_n, h)$$

$$H(q_1, \dots, q_n, -K(q_1, \dots, q_n, p_2, \dots, p_n, h), p_2, \dots, p_n) \equiv h$$

$$\frac{d}{dq_i} \Big| 0 = \frac{\partial H}{\partial q_i} - \frac{\partial H}{\partial p_1} \frac{\partial K}{\partial q_i}$$

$$\frac{d}{dp_i} \Big| 0 = \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_1} \frac{\partial K}{\partial p_i}$$

$$i = \overline{2, n}$$

$$\frac{dq_i}{dq_1} = \frac{dq_i/dt}{dq_1/dt} = \frac{\partial H / \partial p_i}{\partial H / \partial p_1} = \frac{\partial K}{\partial p_i}$$

$$\frac{dp_i}{dq_1} = \frac{dp_i/dt}{dq_1/dt} = \frac{-\partial H / \partial q_i}{\partial H / \partial p_1} = -\frac{\partial K}{\partial q_i}$$

$$\begin{cases} \frac{dq_i}{dq_1} = \frac{\partial K}{\partial p_i} \\ \frac{dp_i}{dq_1} = -\frac{\partial K}{\partial q_i} \end{cases}$$

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 $i = \overline{2, n}$

$$\Rightarrow q_i = q_i(c_1, \dots, c_{2n-2}, q_1)$$

(***)

$$p_i = p_i(c_1, \dots, c_{2n-2}, q_1)$$

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = h(q_1, \underbrace{q_2, \dots, q_n}_{***}, \underbrace{p_1, p_2, \dots, p_n}_{***}) = \tilde{h}(q_1, c_1, \dots, c_{2n-2}, h)$$

K!

$$\int \frac{dq_1}{\tilde{h}(q_1, c_1, \dots, h)} = dt \Rightarrow q_1 = q_1(c_1, \dots, c_{2n-2}, c_{2n}, t)$$

$$p_1 = -K(\underbrace{q_1, q_2, \dots, q_n}_{***}, p_2, \dots, p_n, h)$$